

Radioactivity

Types: Alpha radiation - a release of particles, one of which is the **alpha particle**
 "α"
 $\rightarrow \frac{4}{2}\text{He}$ is the only alpha particle released

X $\rightarrow \alpha$
 X $\rightarrow \gamma$

Beta radiation - a release of particles, one of which is the **beta particle**
 "β"

X $\rightarrow \beta$
 X $\rightarrow \gamma$

$\rightarrow -1e$, electron
 $\rightarrow +1e$, anti-electron/positron

} possible beta particles released

← antimatter of electron ← we (as matter) can't see antimatter

Gamma radiation - a release of gamma electromagnetic waves (high energy photons)
 "γ"

X $\rightarrow \gamma$
 X $\rightarrow \gamma$

↳ highest frequency waves (come from nucleus)

All decays are random and spontaneous

↓
 in a sample of species, any particle can decay

↓
 can happen any time

↳ These decays follow **conservation of energy**

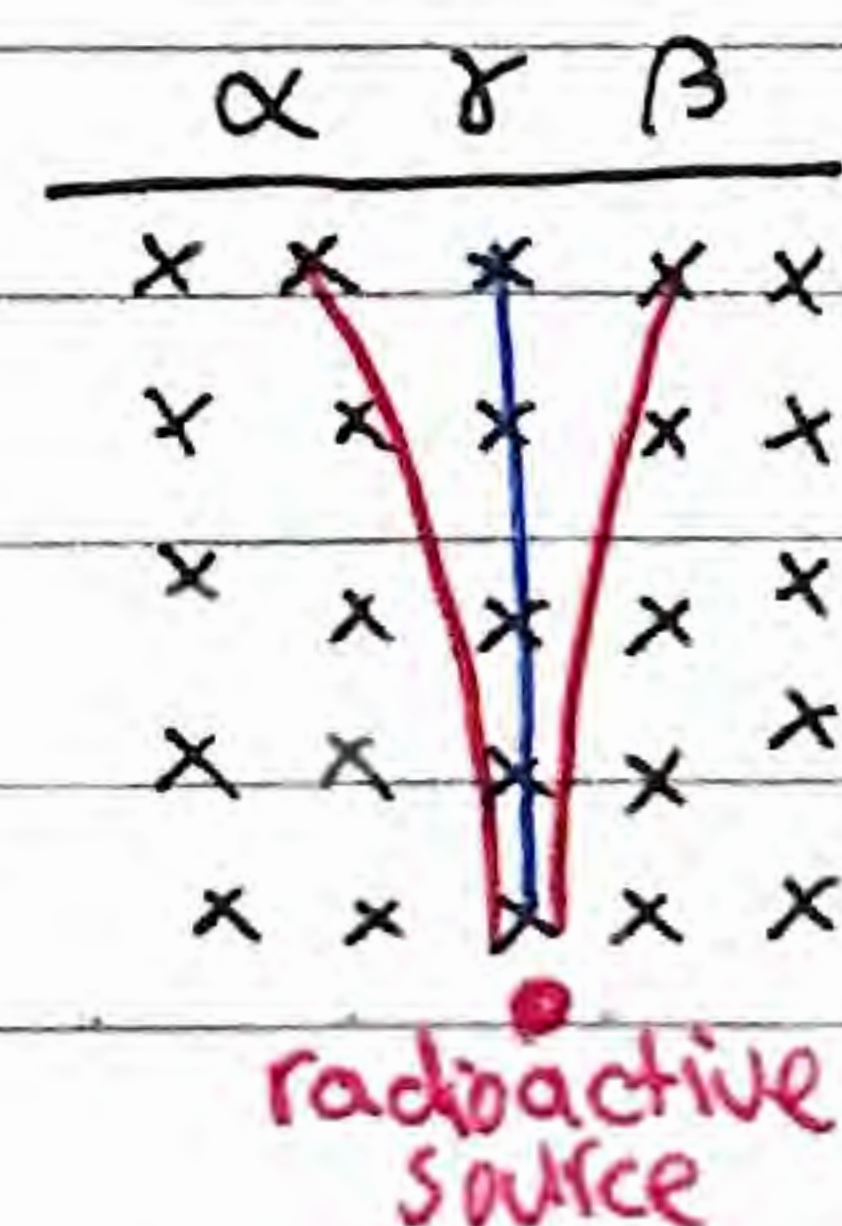
Distinguishing types of radiation

• You can distinguish radiation due to charge using a magnetic field

↳ Gamma (neutral, no deflection)

↳ Alpha & Beta-positron (positive, left)

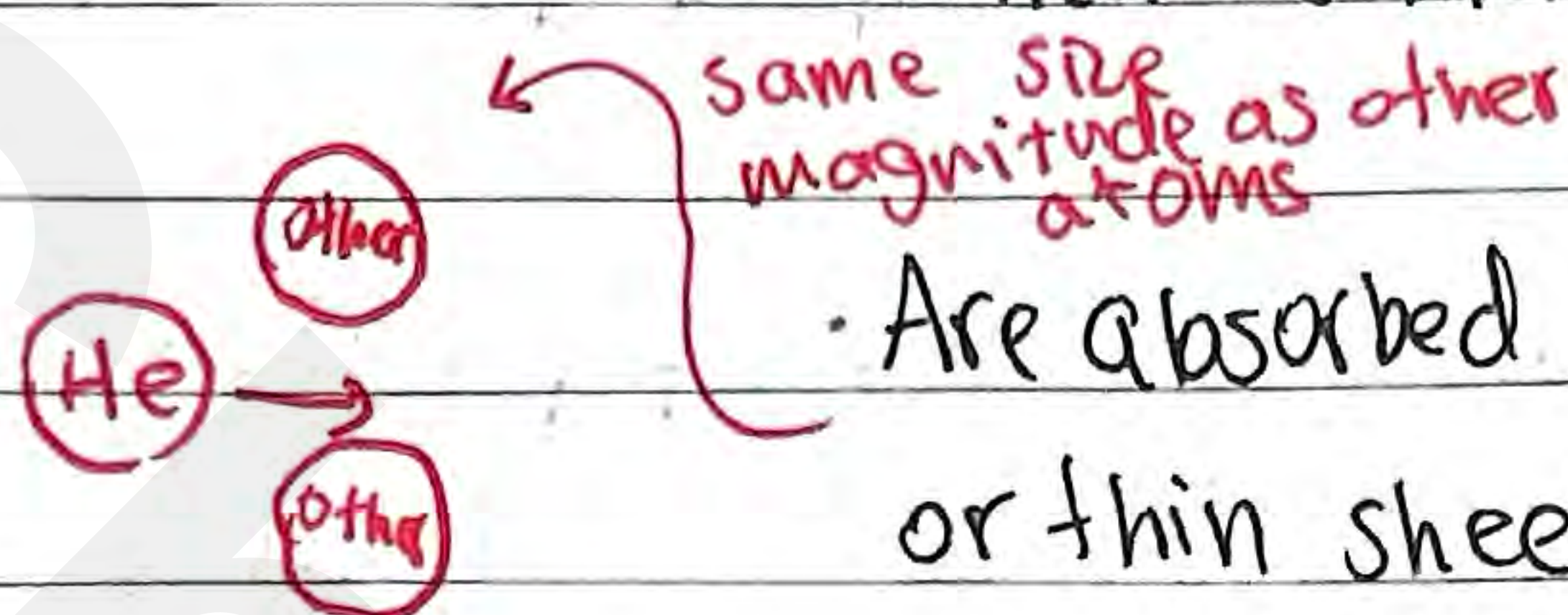
↳ Beta-electron (negative, right)



Radioactivity

Alpha particles :

- Highly ionizing (due to its $2+$ charge)
- Penetrate matter poorly (large size due to being an ion)



- Are absorbed by a few cm of air or thin sheet of paper

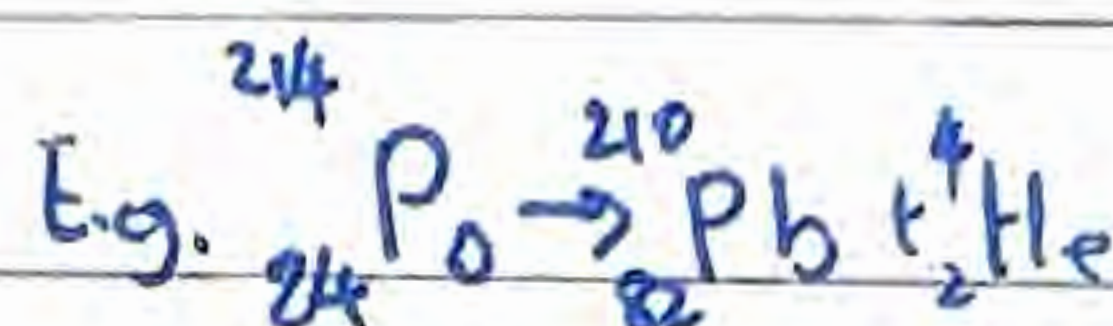
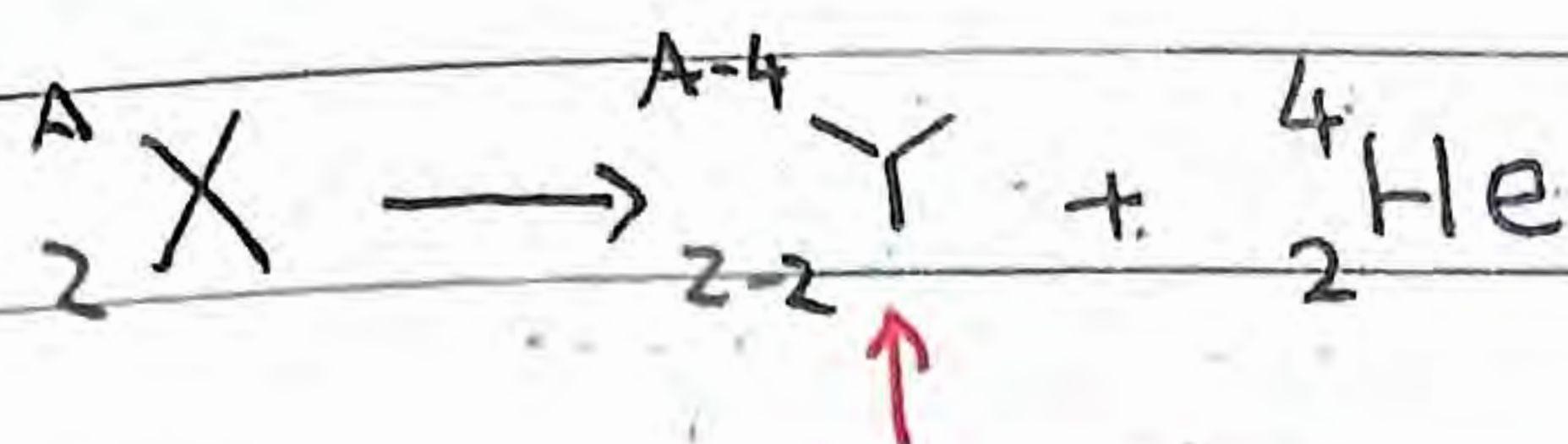
Beta-minus particles :

- Moderate ionizing power compared with alpha particles of the same energy
- Moderate ability to penetrate matter
- Absorbed by ≈ 25 cm air
- Absorbed by few cm of animal/plant tissue
- Absorbed by few mm of aluminum

Gamma photons :

- Weakly ionizing
- Eject electrons from metals (photoelectric effect), causing secondary ionization
- Highly penetrating
- Absorbed by few cm of lead
- Absorbed by few m of concrete
- Only weakly absorbed by animal tissue

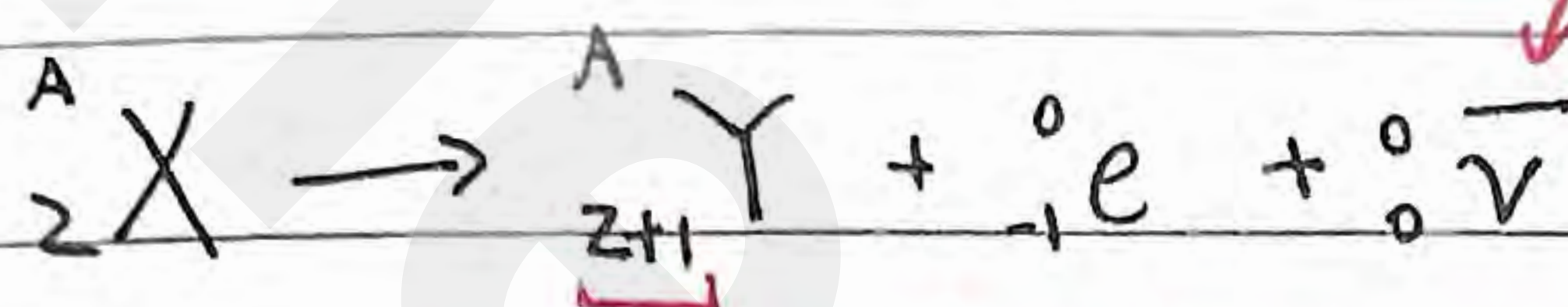
Alpha Decay



find element Y from atomic #

Beta-minus Decay

↑
occurs in
neutron-rich
nucleides
(ratio between
neutrons
protons)



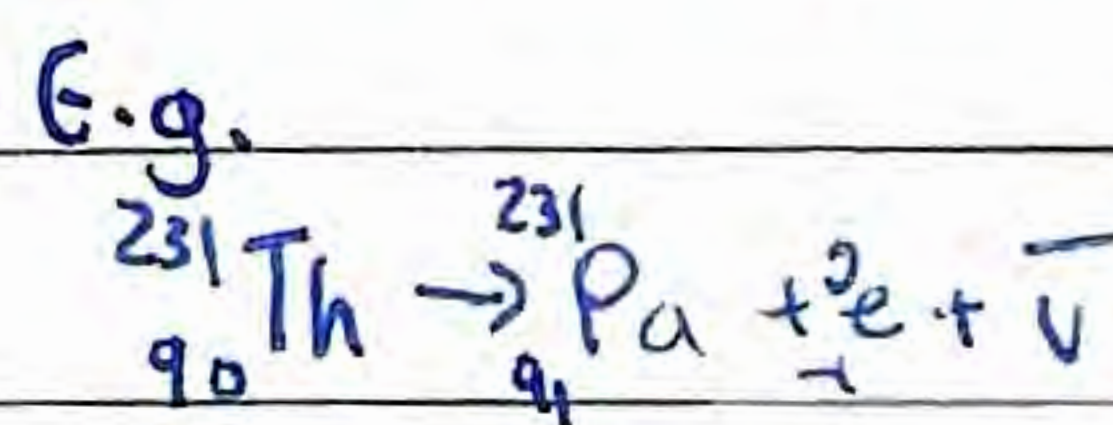
↓ $\bar{a}$ denotes anti-
anti-neutrino
(energy difference between
product and reaction)
↳ To satisfy energy conservation

where did proton come from?

↳ protons are not the smallest indivisible particle

↑
tries to decrease
neutron-richness

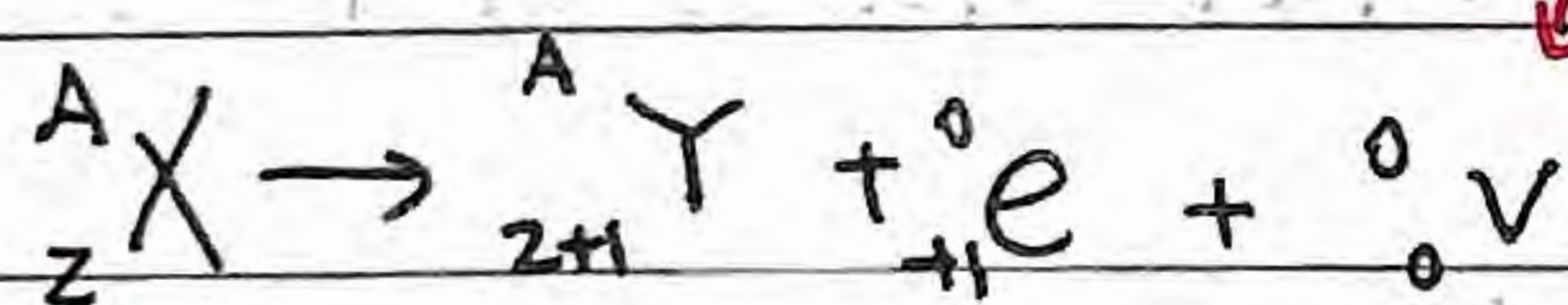
↳ neutron $\rightarrow$ proton + e
made of quarks



Beta-plus Decay

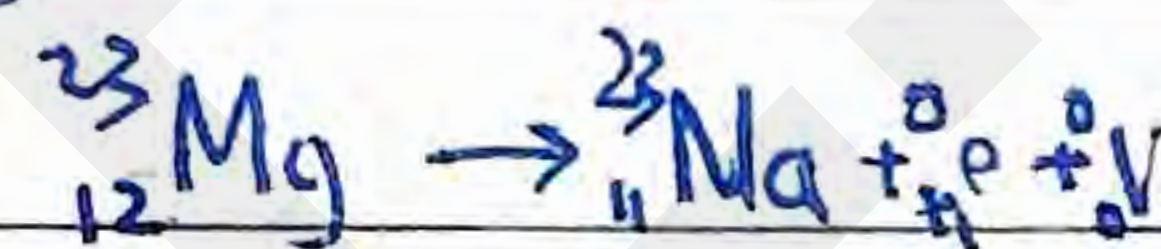
↑
occurs in
proton-rich
nucleides

↑
tries to decrease
proton-richness



↳ neutrino (to balance energy)

E.g.

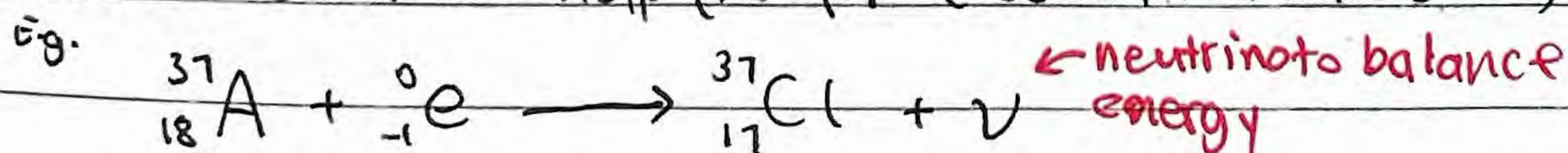


proton $\rightarrow$ neutron + ${}^0_{+1} e$

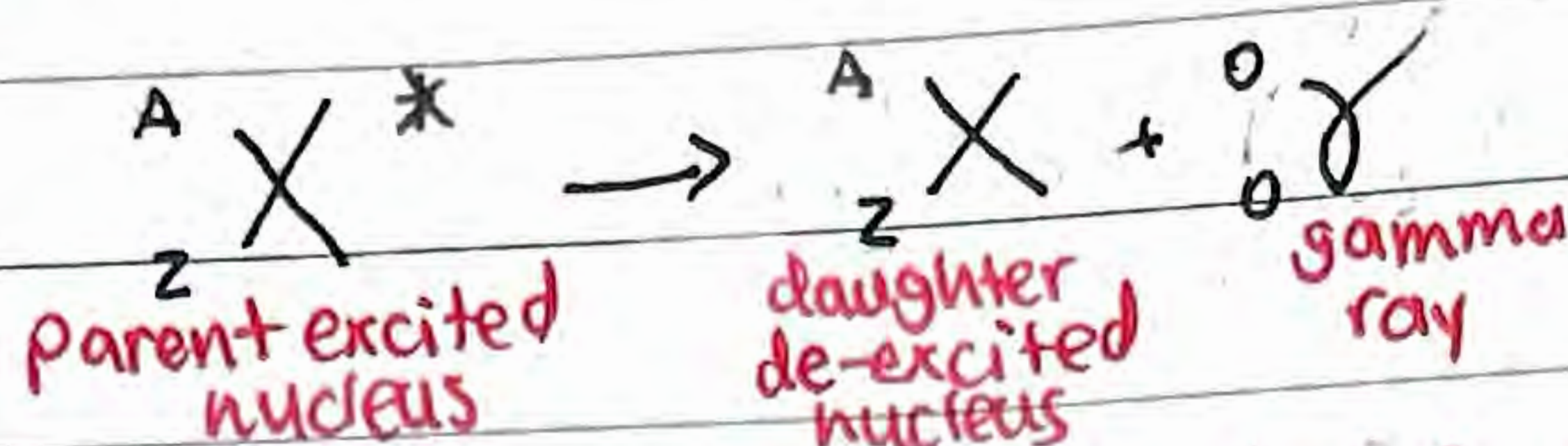
Electron-Capture Nuclear Reaction

An alternative way in which proton-rich nucleides can increase the neutron/proton ratio. "artificial transmutation"

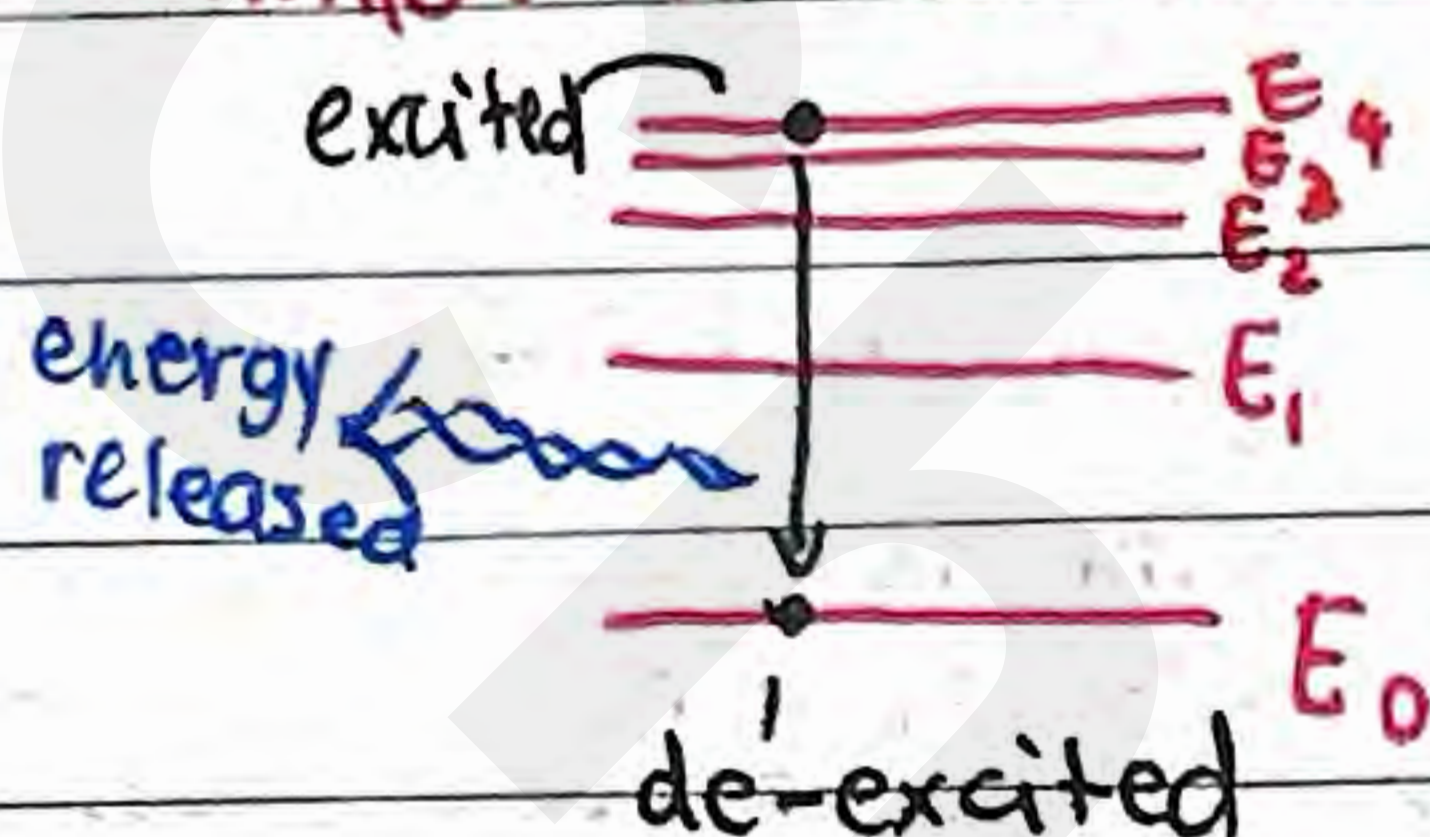
Requires external help (i.e. for electron to hit nucleus)



Gamma Decay



Similar to electron energy levels, but for nucleus



will be said in question

↳ Typically follows from alpha/beta decays

↳ The product can sometimes be excited by which its unstable, thus gamma decay

The Radioactive Decay Law

The rate of decay is proportional to the number of nuclei present that have not decayed yet.

$$\text{rate of decay} = \frac{dN}{dt} \propto -N = \# \text{ of undecayed neutrons}$$

(negative since rate of change of undecayed nuclei) logistic differential equation

The more undecayed neutrons, the greater the rate of decay

$$\frac{dN}{dt} = -\lambda N$$

decay constant Per each isotope

The Radioactive Decay Law

Solving the differential equation...

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt$$

$\rightarrow$ initial (total nuclides)

$$\ln(N) \Big|_{N_0}^N = -\lambda t \Big|_0^t \Rightarrow \ln\left(\frac{N}{N_0}\right) = -\lambda t$$

Thus...

$$N = N_0 e^{-\lambda t}$$

of nuclei at a given time t that haven't decayed yet

But how about # of nuclei that have decayed?

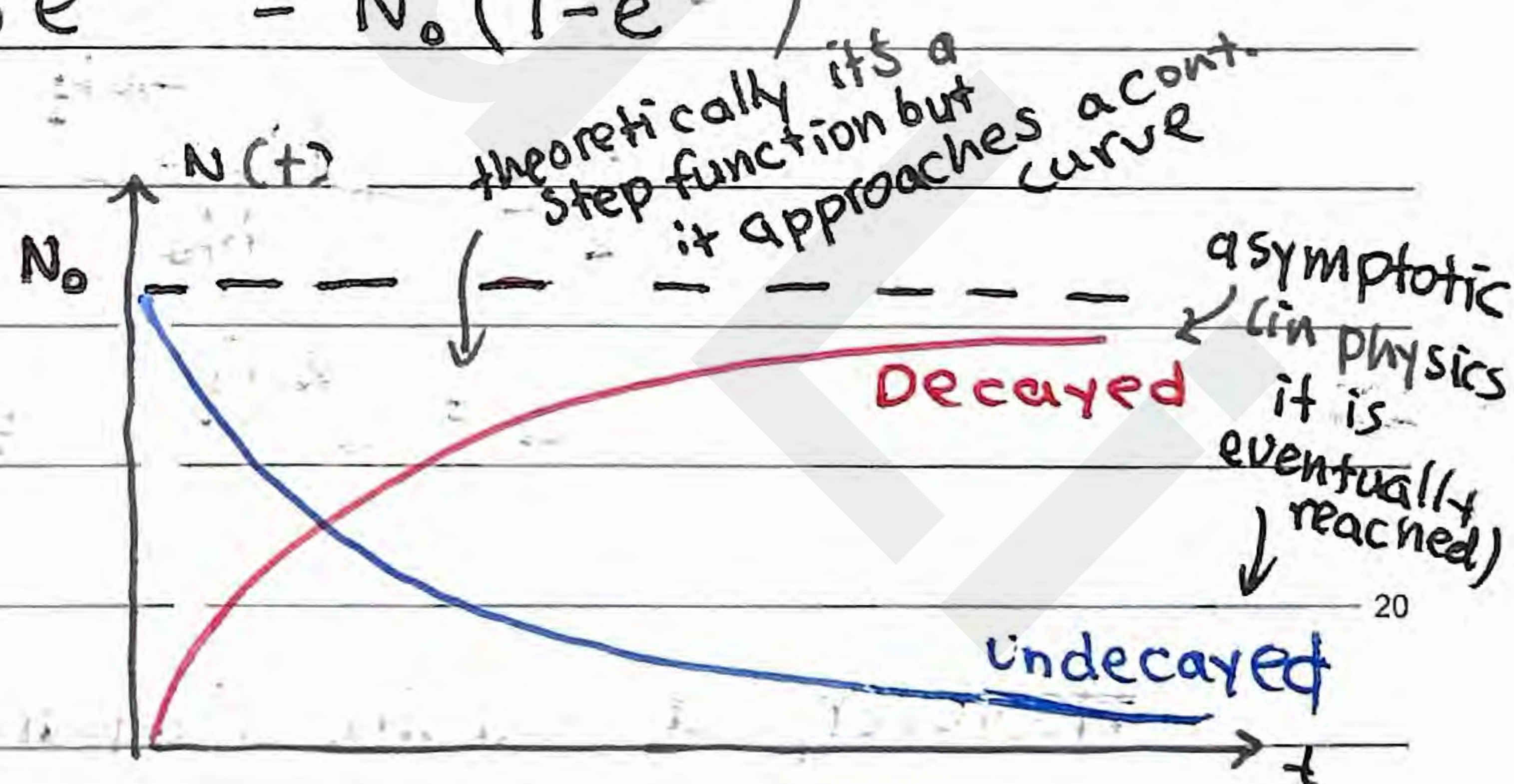
$$N_0 - N = \text{Total} - \text{Undecayed} = \text{Decayed}$$

$$\therefore \text{Decayed} = N_0 - N_0 e^{-\lambda t} = N_0 (1 - e^{-\lambda t})$$

Thus (on a graph)...

$$\bullet \text{ Decayed} = N_0 - N_0 e^{-\lambda t}$$

$$\bullet \text{ Undecayed} = N_0 e^{-\lambda t}$$



Another word for the rate of decay is **Activity**

$$A = -\frac{dN}{dt} = \lambda N = \lambda N_0 e^{-\lambda t} = \underbrace{\lambda N_0}_{\text{group}} e^{-\lambda t} = \overset{\text{initial activity}}{A_0} e^{-\lambda t}$$

The unit is # of nuclei per second or Becquerel (Bq)

Radioactive decay

The decay constant is constant for each isotope:

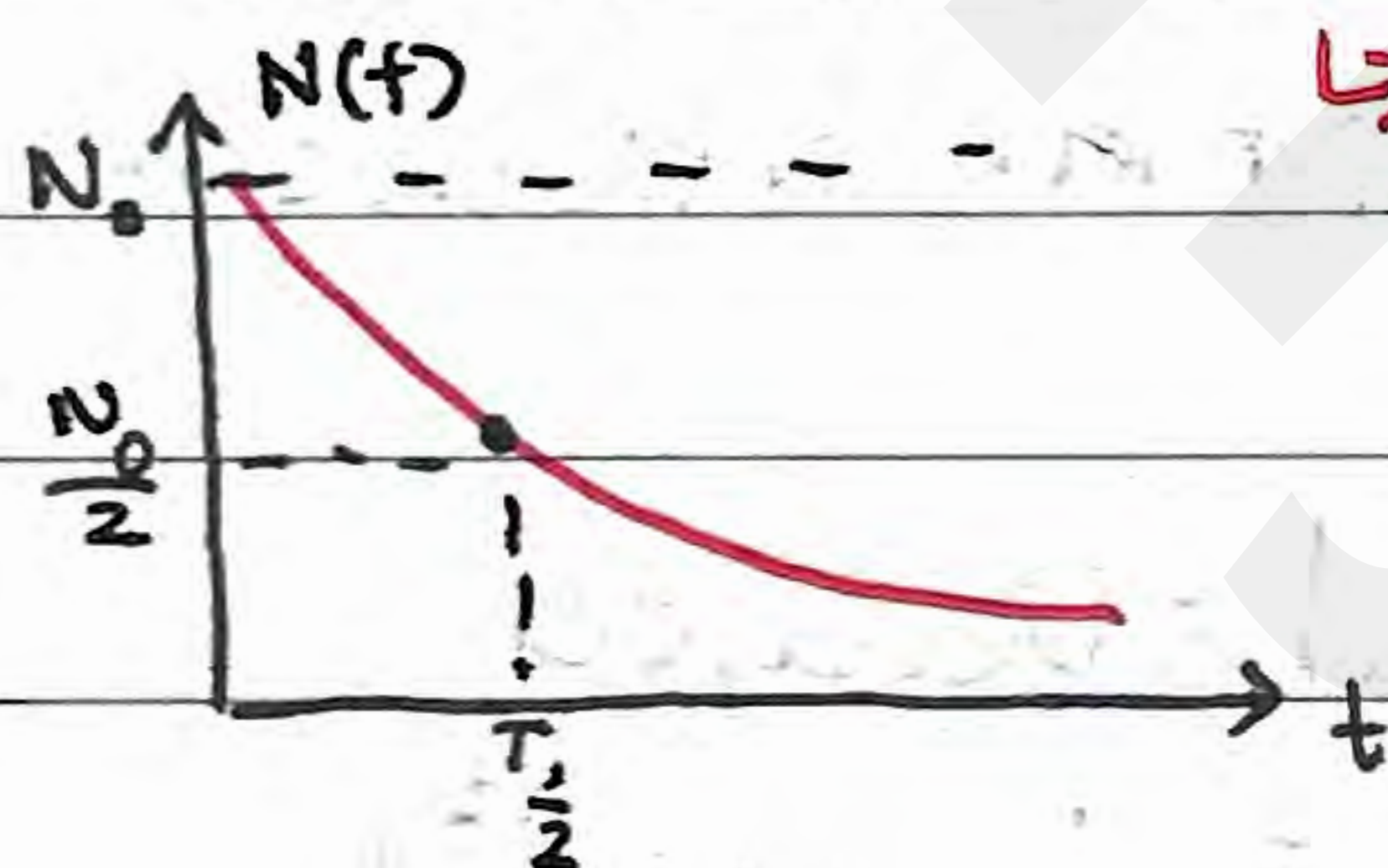
$$\lambda = \frac{\text{probability of decay}}{dt}$$

This constant connects with half-life

$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$$

↳ How much time has passed for half of the initial nuclei (N_0) to decay?

↳ It is also time for Activity to half ($A \propto N$)



→ $2T_{1/2}$ is when N_0 decays to $\frac{N_0}{4}$

$3T_{1/2}$ is when N_0 decays to $\frac{N_0}{8}$

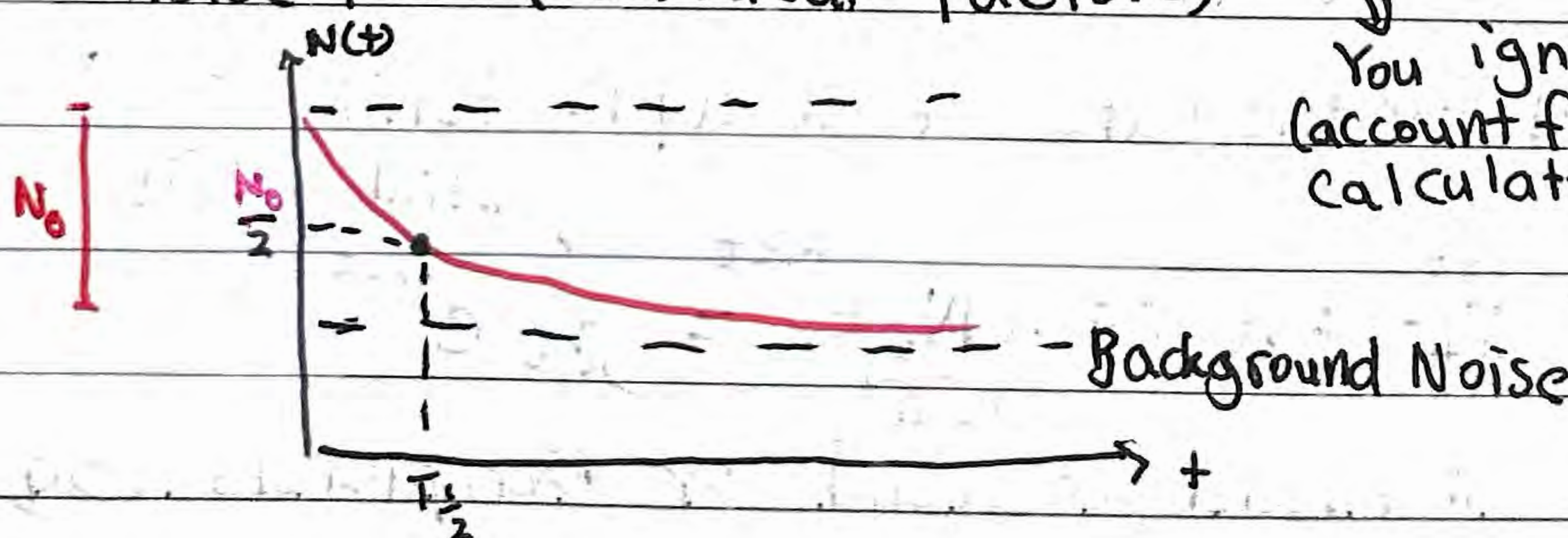
(keep halving)

$$\frac{1}{2} = \frac{N_0 e^{-\lambda T_{1/2}}}{N_0}$$

$$2 = e^{\lambda T_{1/2}} \Rightarrow T_{1/2} = \frac{\ln 2}{\lambda}$$

↳ useful to determine λ from half-life

However, a real-life measured decay graph has background noise too (unideal factors)



You ignore that (account for it) in calculating half-life

Radioactive Decay

- For time periods, typically there is no need for a conversion of units

↳ When given ($T_{1/2}$) is mins, then your result (t) is also mins

- $\lambda \cdot N_0 = A_0$ (often expressed as decay rate before event observed)

Background Radiation (Noise)

- When a radiation detector is left without a prominent source of a radiation, it will still detect ionizing radiation
- The most typical type of detector is Geiger counter
- When a problem asks for final count after time, make sure to subtract background radiation, calculate, and add back the background radiation

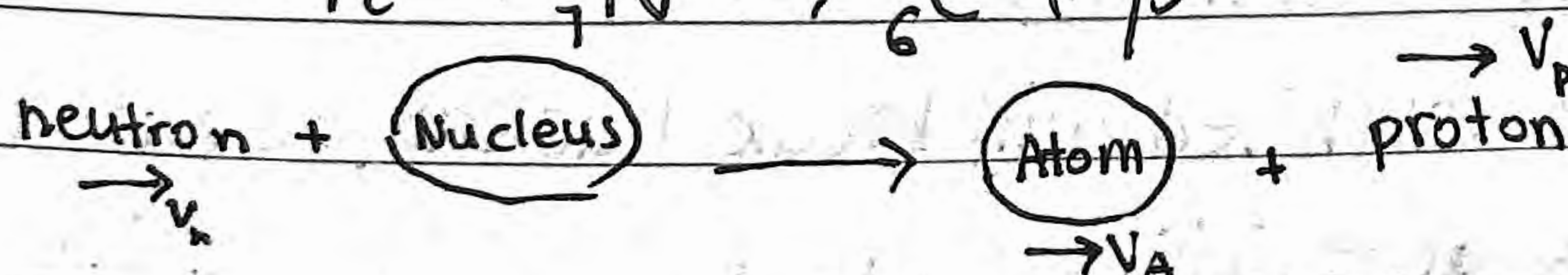
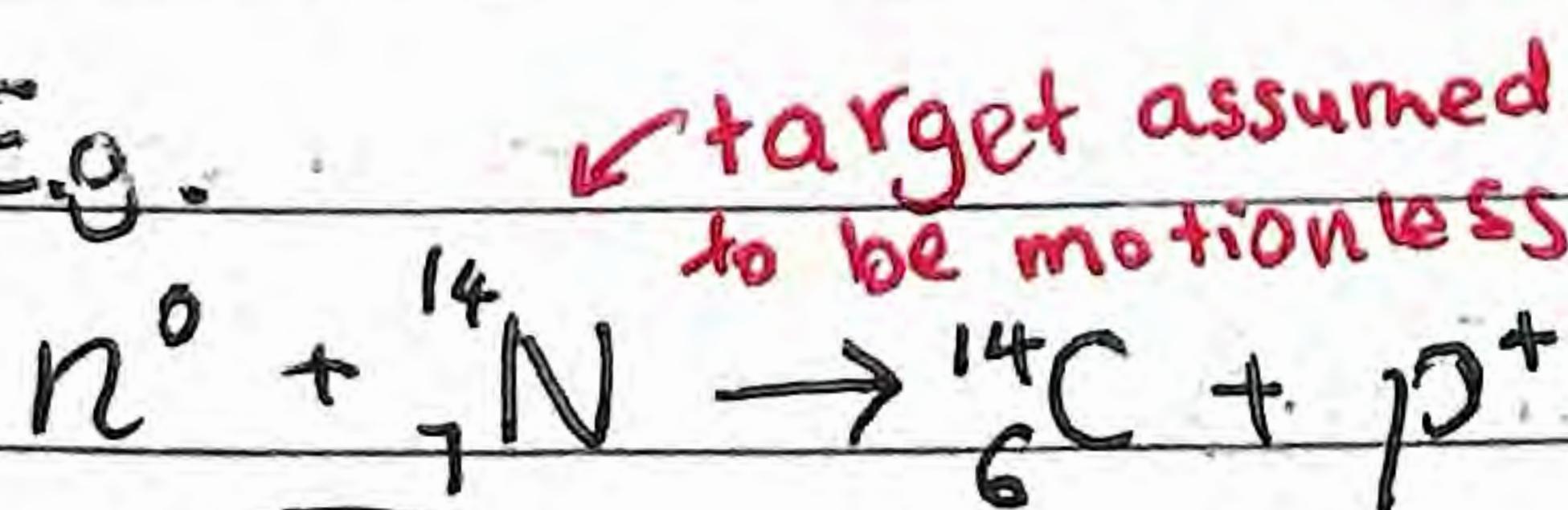
Nuclear Reactions

- Transmutation - Transformation of one element into another

Can be both natural and artificial

↳ Occurs when a nucleus is struck by a nucleus, neutron or γ ray

↳ Reaction E.g.



Nuclear Reactions

• These reactions must follow conservation of particles

↳ where $\#_{init} = \#_{final}$; $\#_{i,n^0} + \#_{i,p^+} \rightarrow \#_{f,n^0} + \#_{f,p^+}$

↳ Thus, $\#$ nucleons is conserved, $\#$ protons is conserved
($\#$ neutrons is implied) $\rightarrow$ rel. atomic mass

Reaction Energy (reactions must have energy)

• Since mass conservation is not guaranteed even with same total $\#$ of protons and neutrons, we can use mass difference for energy change

For general... $a + X \rightarrow Y + b \dots$

$$Q = \Delta E, \quad Q = (BE_Y + BE_b) - (BE_a + BE_X) = \Delta \sum BE$$

alternatively (via mass difference...)

$$E = mc^2 \rightarrow Q = -\Delta m \cdot c^2 \leftarrow \text{generalized for more terms}$$

$$= [(m_a + m_X) - (m_Y + m_b)] c^2$$

↳ Typically, mass difference is used to calculate binding energy

Just as in regular reactions...

$Q > 0$, exothermic (energy released)

↳ mass decrease

$Q < 0$, endothermic (energy absorbed)

↳ mass increase

Nuclear chain reactions

↳ A particle launched will break nucleus, release more launched particles that initiate more reactions (breaking nucleus releases energy)

Nuclear Reactions

E.g. Uranium



For initial success,
n must collide perfectly
at center of mass ${}^{235}\text{U}$



Continue on
in a chain

After that (depending on
speed of n_{init}), A and
B are formed (not necessarily
same element in every scenario)

large nuclei is decreasing segment of BE

Fission - Large nuclei split apart

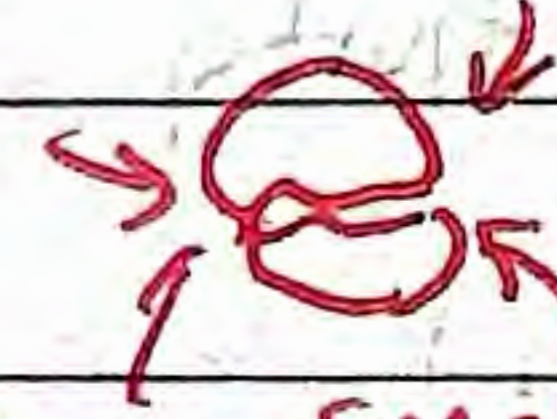
↳ exothermic

Binding Energy, $BE_f > BE_i$

Fusion - Small nuclei join together

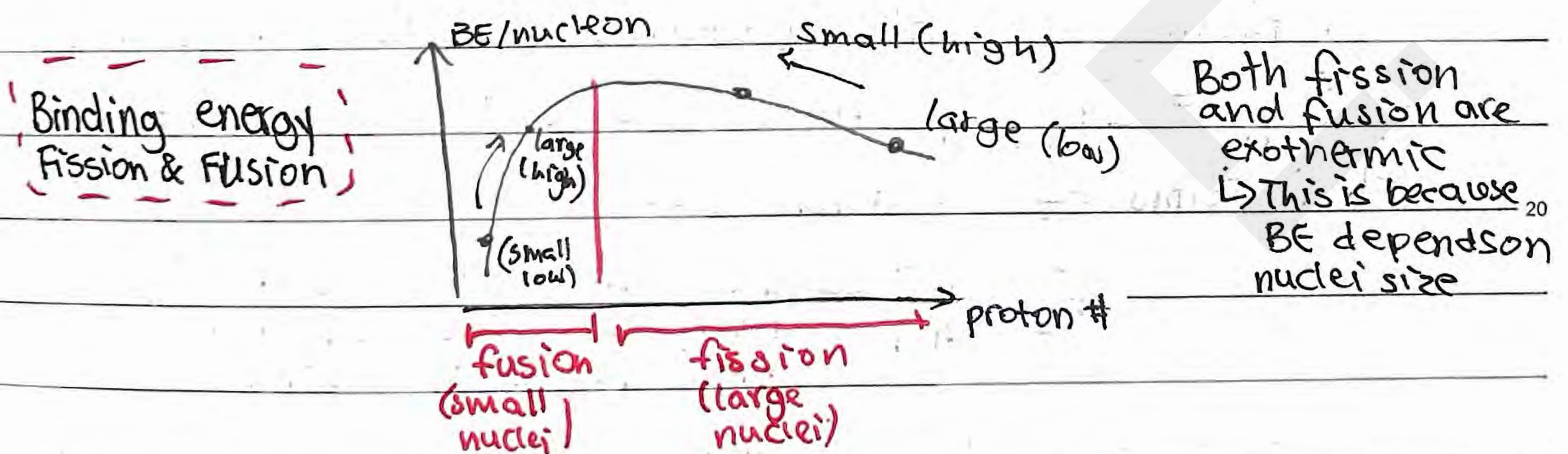
↳ exothermic

$BE_f > BE_i$



*How do we know if a reaction starts?

Find Q of reaction, does initial supply of Q
meet that? → To start, it's endothermic

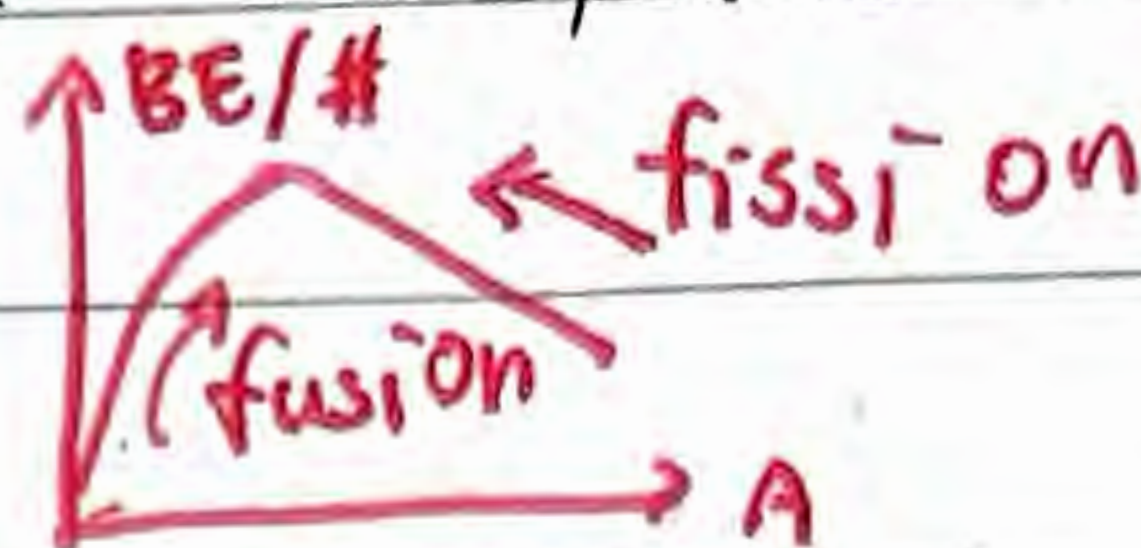


Additionally, heat can be calculated from binding energy graph

↳ Make sure to multiply by # of nucleons!!

Fission - Process in which a heavy nucleus splits up into lighter

↑ nuclei
both are exothermic



Fusion - The joining of two light nuclei into a heavier one
(can occur naturally or induced)

Radioactivity - Emission of particles and energy from a nucleus
↳ Random and spontaneous

Radioactivity and mechanics

↳ Radioactivity connects to momentum and energy

↳ Binding energy is typically ΔE_{tot} .

↳ This is reflected by the ΔE_k (when rest particle decays)

E.g. when $^{226}\text{Ra} \rightarrow ^{222}\text{Rn} + ^4_2\text{He}$, the $BE = \Delta E_{k, \text{tot}}$.

When finding E_k , it's useful to observe momentum

↳ $\sum p_{\text{int}} = \sum p_{\text{final}}$, since $p = m \cdot v$, we can get E_k in terms of p and m ...

$$E_k = \frac{p^2}{2m}$$

← The ratio between E_k of two particles is the ratio between their masses if $|p| = |p|$

↳ This is useful to substitute and simplify ΔE in terms of only one E_k , making $BE = \Delta E_k$ solvable

Radioactivity - 1

· Additionally, remember binding energy is for one reaction - one nucleus

↳ Thus, when given a sample, make sure to obtain how many nuclei there is and multiply for total energy.

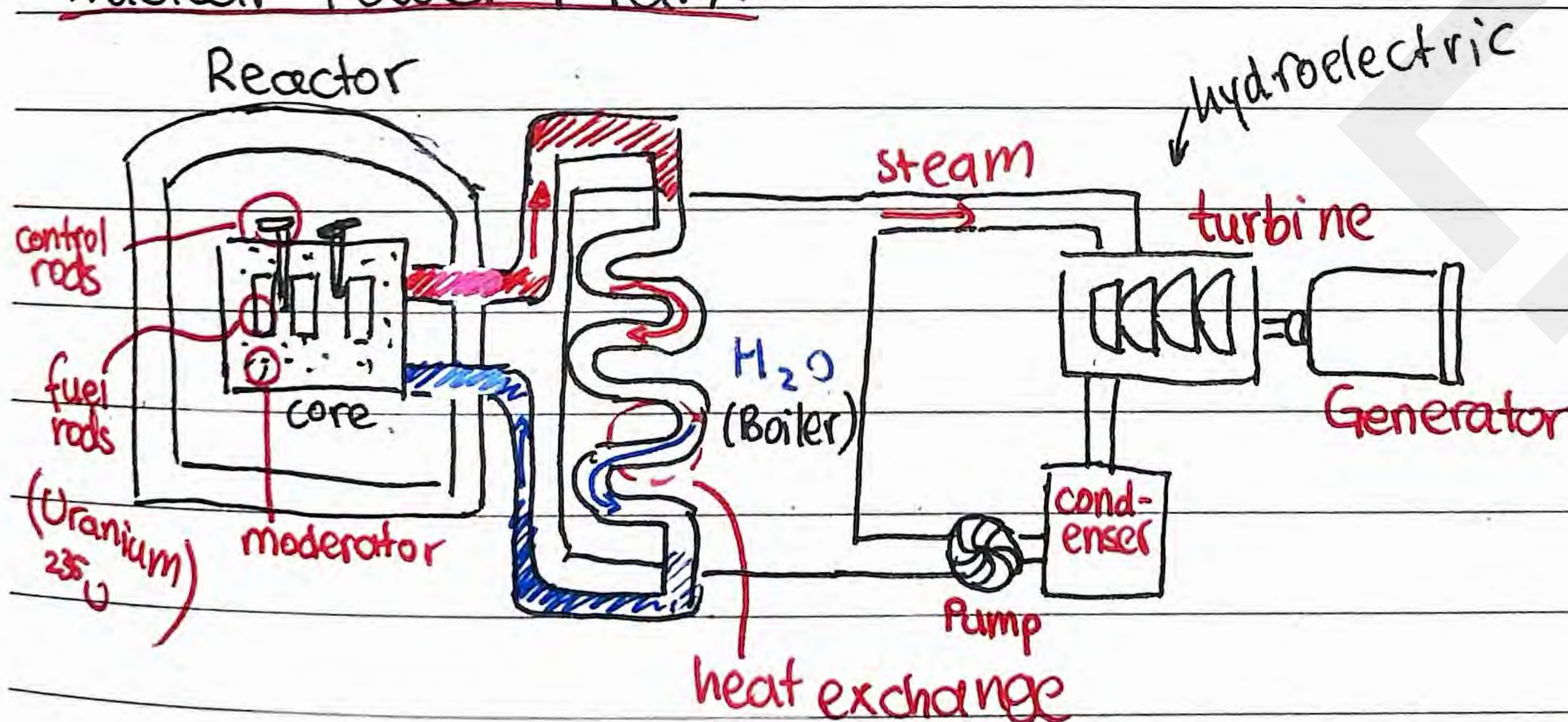
$E = mc^2$ is $E \propto m$

· So this means ratio between masses is ratio between energies → E.g. Fraction of uranium mass to energy of reaction

$$\text{Thus } \frac{\Delta m}{m_0} = \frac{\Delta E_{\text{re}}}{4E_{\text{tot}}}$$

> Spontaneous fission can occur but highly rare

· Induced fission is more common since energy required for reaction to start (even if it's exothermic)

Nuclear Power Plant

Moderator - Slows down the speed of neutrons (made of carbon, water)

Control Rods - Absorbs the neutrons (made of boron)